7. SITUATIONAL INJURIES

The injuries that arise from certain situations in which first aid treatment could be applied such as:

- i. Poisoning
- ii. Drug Overdose
- iii. Bites

i. POISONING

- Poisons are chemical substances used at home, in industries, in schools or agricultural setting which when swallowed or in contact with the body may cause varying degree of irritation and damage to the body functions.
- Examples of poison are: gasoline, soaps, pesticide, insecticide, expired drugs, bleach, detergents etc
- Poisoning is the sudden illness that results from the effect of poison in the body

Routes of Poisons

- Poison gets to the body through the following routes:

i. the mouth: Poison could get to the body through ingestion (eating and drinking) of poisonous substance either in solid or liquid form

ii. the nose and Air passage: this occurs through inhalation of poisonous gases through the nose or mouth into the lungs; gases like carbon monoxide, ammonia are poisonous gases that can impair body function and may lead to death

iii. the skin: Poison could also get into the body through the skin by contact with or absorption of certain chemicals such as pesticides, insecticide injection of expired or poisons from bites, reptiles and animals.

Signs and Symptoms

- Strong smell
- Shallow breathing
- Vomiting
- Burns of the lips and mouth
- Cramping abdominal pain
- Asphyxia
- Convulsion
- Coma

FIRST AID TREATMENT

- Delay of any sort should be avoided, If conscious
- Induce casualty to vomit
- Give the casualty lot of water/milk to drink
- Send the casualty to the hospital
- **If unconscious**
- Lay the casualty in a recovery position,
- Clear air passage and loose all constriction
- Apply artificial ventilation to restore breathing
- Send the casualty to the hospital

Prevention of Poisoning

- Keep all drugs/household chemicals out of reach of children
- Avoid eating rotten fruits
- Avoid taking expired drugs or food items
- Avoid contaminated water or food items
- Avoid fake drugs and food substances

Substances applicable in the First Aid treatment of Poison

- First aid treatment of poisoning could be given through certain substance. These are:
- **Antidotes**: They are substance that neutralize or counteract the effect of poisons. The kind of poison ingested will determine the type antidote to use. Examples of antidotes are: activated charcoal, naloxone, sodium bicarbonate

Substances applicable in the First Aid treatment of Poison

- **Emetics**: these are substance that are capable of inducing vomiting. They are given in order to remove poison from the body before it is absorbed into the blood stream through vomit. Examples are: magnesium sulphate (epsom salts), ipecacuanha, sodium chloride, sodium carbonate (washing soda), mustard

Substances applicable in the First Aid treatment of Poison

- **Demulcents**: They are substances that soothe initial mucous membrane and they are given after the casualty must have vomited freely. Demulcents are commonly used to soothe irritated throats. Examples are: milk, honey, olive oil, etc

Classification of Poisons

Poisons can be classified in several ways, including by their mode of action and chemical nature. Based on mode of action, they can be categorized as narcotic, corrosives, irritants poisons.

Narcotic Poisons: These are poisons which act on the nervous system and have various effects, such as:

i. Hypnotic effect: this leads to insomnia and unconsciousness
ii. Delirious effect: This initially produces excitement and later leads to unconsciousness.
iii. Convalescent effect: This initially produces twitching of the limb but later leads to unconsciousness

Classification of Poisons

Signs and symptoms

- Slow breathing
- Feeble pulse or feeling of suffocation
- Drowsiness
- Twitching of muscles
- Coma
- Death may occur.

First Aid Treatment

- If the casualty is drowsy, keep the victim awake
- Give emetics
- Give artificial ventilation

Classified Poisons

Irritant Poisons – They acts on the alimentary tract leading to irritation and inflammation on the tract. It is caused by chemical substances or contaminated food.

Signs and symptoms

- Nausea
- Vomiting
- Diarrhea
- Acute abdominal pain

First Aid Treatment

- Give antidote if the actual poison is known
- Give an emetic if the actual poison is unknown or
- Give demulcent after the casualty has vomited
- Seek medical assistance

Classified Poisons ctd

Corrosive Poison: These are poisons which burn any part of the body they come in contact with, it can be acid or alkaline

Signs and Symptoms

- Severe burns around the mouth and lips
- Pains in the mouth, chest, throat and abdomen.

First Aid Treatment

- Give an antidote
- Give demulcent to soothe the damaged mucous membrane
- Treat for shock
- Seek medical assistance

ii. Drug Overdose

- Drug overdose occurs when the drug is ingested or injected in an amount above and more than the set safety limits. It may occur by mistake, out of ignorance or a suicide attempt.

First Aid Treatment

If Conscious

- Induce vomit
- Give mild laxatives
- Give plenty of water

If Unconscious

- Do not induce and do not give anything to eat or drink
- Lie on the side on the ground and perform artificial resuscitation if necessary.

ii. Drug Overdose

Prevention of Drug Overdose

- Keep all drugs out of reach of children
- Ensure only prescribed drugs are taken with the prescribed dosage.

iii. Bites

- A bite is a cut made on the skin with teeth by an insect or animal through which toxin or venom (poison) is deposited with various degrees of effect on the body.

Types of Bites

- Insect bite
- Animal bite

Insect Bite: These are bites by insects in the form of stings through which toxin is deposited into the body. The common insects that bite or sting are: mosquitoes, bees, tsetse fly, scorpion, etc

- The toxin deposited may cause itching and swelling, dizziness, shivering, cold, etc.

iii. Bites

Insect Bite

Dog Bite

Snake Bite

Bites CONTD.

First Aid Treatment

- Lay the casualty in a comfortable position
- Apply menthol ointment or an antiallergic cream
- Tie a tourniquet a little above the sting for a short period
- Apply cold compression on the sting bud
- Give pain-relieving tablets
- Seek medical assistance if necessary

Bites CONTD.

- **Animal bites:** These are usually dangerous and require prompt attention. Animal bites can be from dogs, snakes, cats, birds, humans, etc, with various degrees of poison. The most common animal bites are from dogs and snakes
- Bites by any of these animals are dangerous and require prompt assistance. A dog bite could be infected with rabies, which could cause death.

First Aid Treatment

- Lay the casualty in a comfortable resting position
- Stop bleeding if any
- Treat as an infected wound.

Bites CONTD.

Snake bite leads to deposit of a poisonous substance called venom into the body and the effect of it depends on the type of snake.

There are two types of snake:

i. Poisonous (venomous) snakes
ii. Nonpoisonous (non venomous) snakes

- The difference in the two types of snake bite is shown at the bite marks. The bite of poisonous snake leaves marks of the two fangs. Nonpoisonous leaves two rows of the marks.

- The effect of poisonous snakes like python or cobra could lead to death. The effect of venom of poisonous snakes start to manifest within thirty minutes and would last over several days.

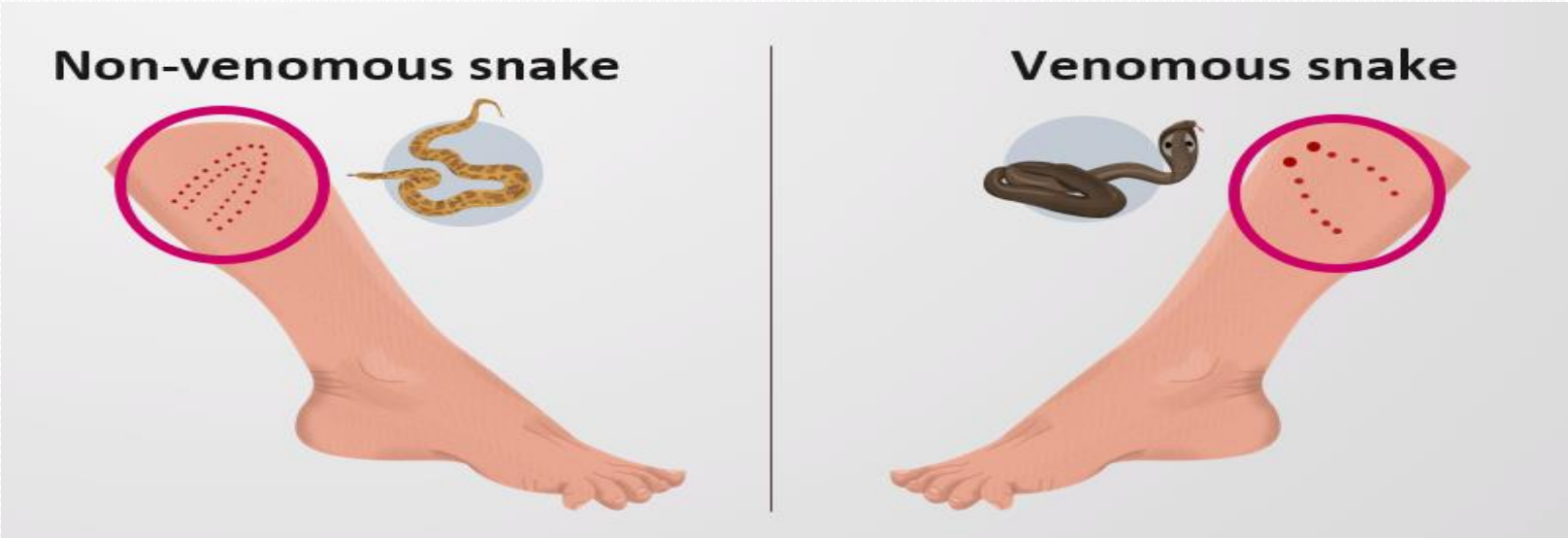
Bites CONTD.

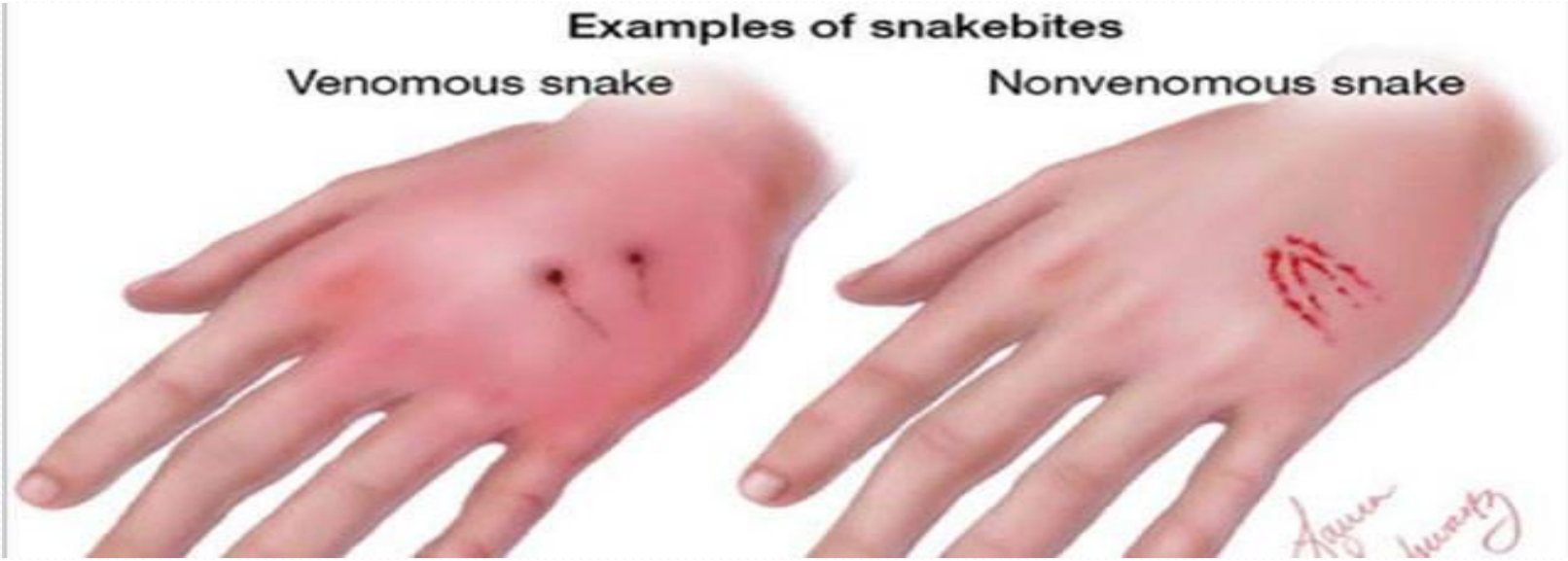
- **Signs and Symptoms**
- Puncture marks may be seen
- Swollen and redness around the bitten point
- Pain and bleeding around the bitten point
- Vomiting may occur
- Disruption of the breathing mechanism
- Disruption of heart and nervous functions
- There could be paralysis

First Aid Treatment

- Lay the casualty down in a quiet place and immobilize the affected part
- Tie a tourniquet or a piece of cloth just above the point of bite
- Observe the casualty and perform emergency resuscitation if necessary.
- Seek medical assistance immediately

Bites CONTD.


Non-venomous snake
Venomous snake


Examples of snakebites
Venomous snake
Nonvenomous snake

8. DRESSING AND BANDAGING

An important aspect of first aid treatment is the application of dressing and bandaging.

DRESSING

- Dressings are applied to cover the wound to control bleeding and prevent infection of the wound.
- Application of dressing requires that:

  i. It is directly on the wound,

  ii. The dressing pad extends beyond the wound,

  iii. More pad should be applied if blood seeps

  iv. Use disposable gloves if available or wash hands before applying dressing

  v. Avoid touching the wound or any part of the dressing that will come into contact with the wound

FORMS OF DRESSING

- There are different forms of dressing, which include:
- **Sterile Dressing**: It consists of a dressing pad attached to a roller bandage. The pad is a piece of lint /gauze backed by a layer of cotton wool.
- Primarily used for covering surgical wounds, burns, or any wound that requires a sterile environment to prevent infection.

FORMS OF DRESSING

- **Gauze Dressing**: A gauze dressing is a type of dressing made from woven or non-woven gauze, which may or may not be sterile. Commonly used for cleaning, packing, and covering various types of wounds.
- It is mostly suitable for minor wounds, abrasions, or as a secondary dressing over a primary wound dressing.

FORMS OF DRESSING

- **Adhesive Dressing/Plaster**: An adhesive dressing is a type of wound dressing that adheres to the skin, typically around the edges, to stay securely in place over a wound.
- It consists of a gauze or cellulose pad with adhesive backing. It is useful for small wounds.

FORMS OF DRESSING

- **Improvised Dressing**: It involves using any clean material to cover the wound. It is used when a pre-packed dressing is not available

ii. BANDAGE

- Bandages are pieces of light material used to support injuries. They are made of fairly light materials like cotton, linen, gauge, calico, and flannel that are capable of producing warmth.
- In a condition that requires firm/very firm support, elastic and rubber materials/Plaster of Paris (POP) materials are used

Bandage

Plaster of Paris (POP)

ii. BANDAGE

- The purposes are to:
- Immobilise
- Control bleeding
- Reduce swelling, and
- Hold dressing

- General rules include:
- The casualty is in a comfortable position,
- Stay in front of the casualty,
- Apply a bandage to the body's contour
- Apply the bandage firmly
- Avoid fingers and toes, and use a reef knot to tie.

TYPES OF BANDAGE

- Bandages come in various types and forms. The two most common types of bandages in first aid are: Triangular bandage and Roller bandage
- **Triangular Bandage:** A triangular bandage is a versatile piece of cloth, usually made from cotton or muslin, shaped like a right-angled triangle. It has three parts – the point, the base and the two ends
- It can be used as a sling, a broad bandage, a narrow bandage, ring pad
- It is used to secure a splint, keep dressing in position, and to immobilise and support limbs.

TYPES OF BANDAGE

- **Triangular Bandages**

TYPES OF BANDAGE

- **Roller Bandage:** is a long, narrow strip of gauze that is unrolled and wrapped around dressings or joints to secure them in place and provide light compression. It is produced from gauze, cotton, cloth and elastic wrapping materials. It has different width and length, Finger – 2.5cmx5.5m; Hand – 5cmx9.12m; Arm – 10cmx9.12m; Leg – 15cmx9.12m

Applying the Roller Bandage

- When partly unrolled, the remaining roll is called the head/cylinder drum, and the unrolled part is the tail
- Support the part to be bandaged
- Place the tail below the injury, make two straight turns to hold in place
- Make many turns, winding from inside to the outside part of the injured part, each turn should cover between a half and two-thirds of the previous layer of bandaging
- Finish the bandaging with one straight turn and secure with a safety pin

SPECIAL BANDAGES

- **Tubular Gauze Bandage**: is a seamless, conforming sleeve of knitted or woven gauze that's shaped like a sock or tube. It is produced from seamless gauze and designed with an applicator. It is to support/hold the light dressing in place

SPECIAL BANDAGES

- **Scultetus Bandage/many-tailed binder** – produced by stitching many strips of material together in the middle, leaving the ends free. It is used for holding a dressing on a limb

SPECIAL BANDAGES

- **Netelast Bandage**: also called an **elasticated tubular net bandage,** is an open-weave, stretchable mesh sleeve designed to hold dressings or splints in place with gentle, uniform pressure.
- It combines the conformability of tubular gauze with the elasticity of crepe bandages.

SPECIAL BANDAGES

- Crepe Bandage: is a woven, stretchable bandage made from cotton or cotton–linen blends with a crinkled “crepe” texture.
- Its elasticity and breathability make it ideal for light to moderate compression and support

Special Bandages CONTD.

- **Sling**: is a simple supportive device used to immobilise and support an injured arm or shoulder. It relieves pain, prevents further damage, and helps maintain correct positioning during healing.
- A sling is also a band of fabric tied around a casualty's neck to support an injured arm. It is a triangular bandage tied around the neck.

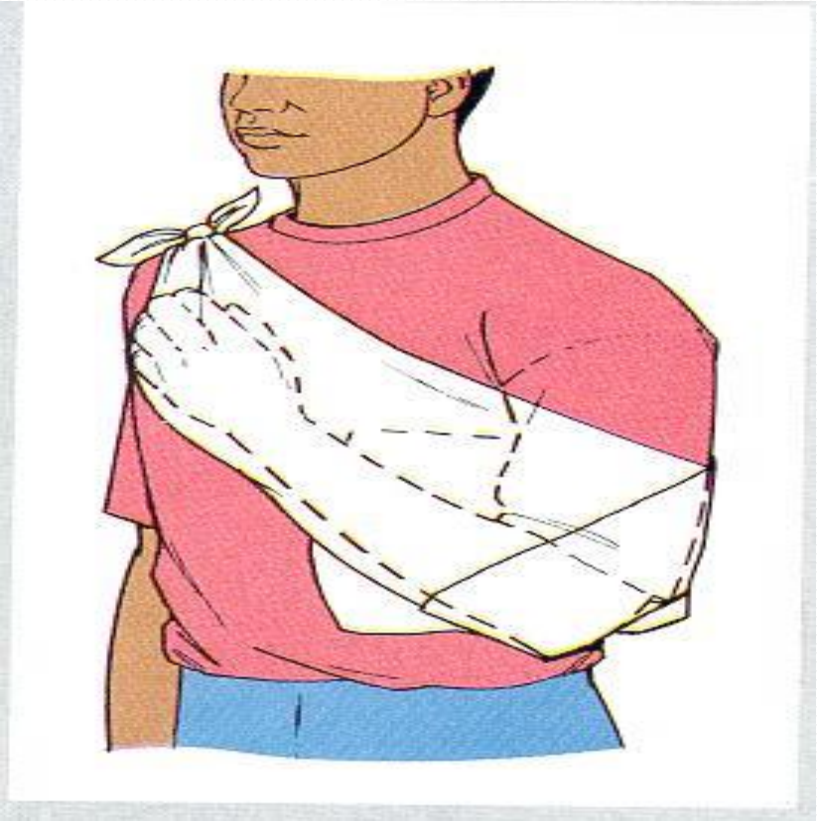
Special Bandages CONTD.

There are two types of sling:

- i. Arm sling
- ii. Elevation sling
- Arm sling supports the forearm, wrist, and hand slightly raised in a horizontal position.
- **Material needed**: A Triangular bandage and a Safety pin.
- **Procedure**
- Starts with the casualty sitting or standing and places the forearm horizontally until the two ends of the bandage are tied in a *reef knot* over the collarbone, and the end is fastened with a safety pin.

Special Bandages contd.

Arm Sling



Elevated Sling

Special bandage contd.

- Elevation Sling: It is a variation of the standard arm sling designed to both support and elevate the injured limb. Keeping the forearm raised above heart level helps reduce swelling and pain, particularly after fractures, soft-tissue injuries, or surgical procedures of the hand, wrist, or forearm.
- Materials needed are: a triangular bandage and a safety pin

Special bandage contd.

- **Procedure**
- The casualty sits or stands and supports the injured arm across the chest with fingertips touching the opposite shoulder.
- Place the triangular bandage over the injured arm such that its base is across the chest and the point hangs loosely
- Tuck the base of the bandage under the hand, forearm and the elbow and bring the lower end up diagonally across the back to meet the other end at the shoulder on the injured side
- Tie the two ends in a reef not at the hollow over the collarbone
- Fold any loose bandage over the point and fasten it with a safety pin

Special bandage contd.

TYING A REEF KNOT

- Pass the left end over and under the right
- Bring both ends up again
- Pass the right end over and under the left
- Pull the ends firmly to tighten, and tuck in the ends

UNTYING A REEF KNOT

- Pull one end and one piece of bandage apart
- Hold the knot, pull the end through it and out

Special bandage contd.

- Improvised Slings – this can be done by:
  - A square of cloth
  - Items of clothing
  - Adjustment of clothes – jacket, button-up jacket, long sleeve shirt or short sleeve shirt.